

GBU 2301: Business Statistics

Week 3: Numerical Measures

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Required reading: Anderson et al., Statistics for Business and Economics, 15th ed., Chapter 3

Outline

- Measures of Location
- Measures of Variability
- Distribution Shape and Relative Location
- Detecting Outliers
- Five-Number Summaries and Box Plots
- Measures of Association Between Two Variables

Numerical Measures

A numerical measure calculated from a sample is a sample statistic. A numerical measure calculated from a population is a population parameter.

Sample	Population
Number of observations: n	Number of observations: N
Mean: $\bar{x}$	Mean: μ
Variance: s^2	Variance: σ^2
Standard deviation: s	Standard deviation: σ

A sample statistic used to estimate the corresponding population parameter is a point estimator.

- The distinction concerns the data source, not the calculation alone.

Mean

The mean is the arithmetic average and the balance point of the observed values.

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} \quad \mu = \frac{\sum_{i=1}^N x_i}{N}$$

- Every observation affects the mean. A few unusually large or small values can therefore move it substantially.
- A mean should be reported with the population or sample, variable definition, unit, and period to which it applies.

Observations: 100

Sample mean net sales: \$77.60

	customer	customer_type	items	net_sales
0	1	Regular	1	39.5
1	2	Promotional	1	102.4
2	3	Regular	1	22.5
3	4	Promotional	5	100.4
4	5	Regular	2	54.0
5	6	Regular	1	44.5
6	7	Promotional	2	78.0
7	8	Regular	1	22.5

Median

- The median is the middle value after the observations are arranged in ascending order.
 - With an odd number of observations, it is the value in the middle position.
 - With an even number of observations, it is the average of the two middle values.
- The median divides the ordered observations into two halves.
 - It is resistant to extreme values because their magnitude does not determine the middle position.
- For strongly skewed income, property value, or transaction data, the median may describe a typical observation more effectively than the mean.

Mode

The mode is the value that occurs with the greatest frequency.

A data set may have:

- One mode
- Two modes
- More than two modes
- No mode when no value repeats

Unlike the mean and median, the mode can summarize categorical data. For quantitative data with many distinct values, the modal value can be unstable or uninformative.

Weighted Mean

- A weighted mean gives observations different importance.

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$$

- The weights must represent the application.
 - Credit hours can weight grades, portfolio shares can weight asset returns, and quantities purchased can weight unit prices.
- If portfolio assets have returns of 8, -2, and 5 percent with portfolio weights of 50, 20, and 30 percent, the portfolio return is 5.1 percent.
- The unweighted mean of 3.67 percent answers a different question.

Geometric Mean

The geometric mean of n positive values is

$$G = \left(\prod_{i=1}^n x_i \right)^{1/n}$$

For successive returns, use growth factors:

$$\bar{r}_G = \left[\prod_{t=1}^n (1 + r_t) \right]^{1/n} - 1$$

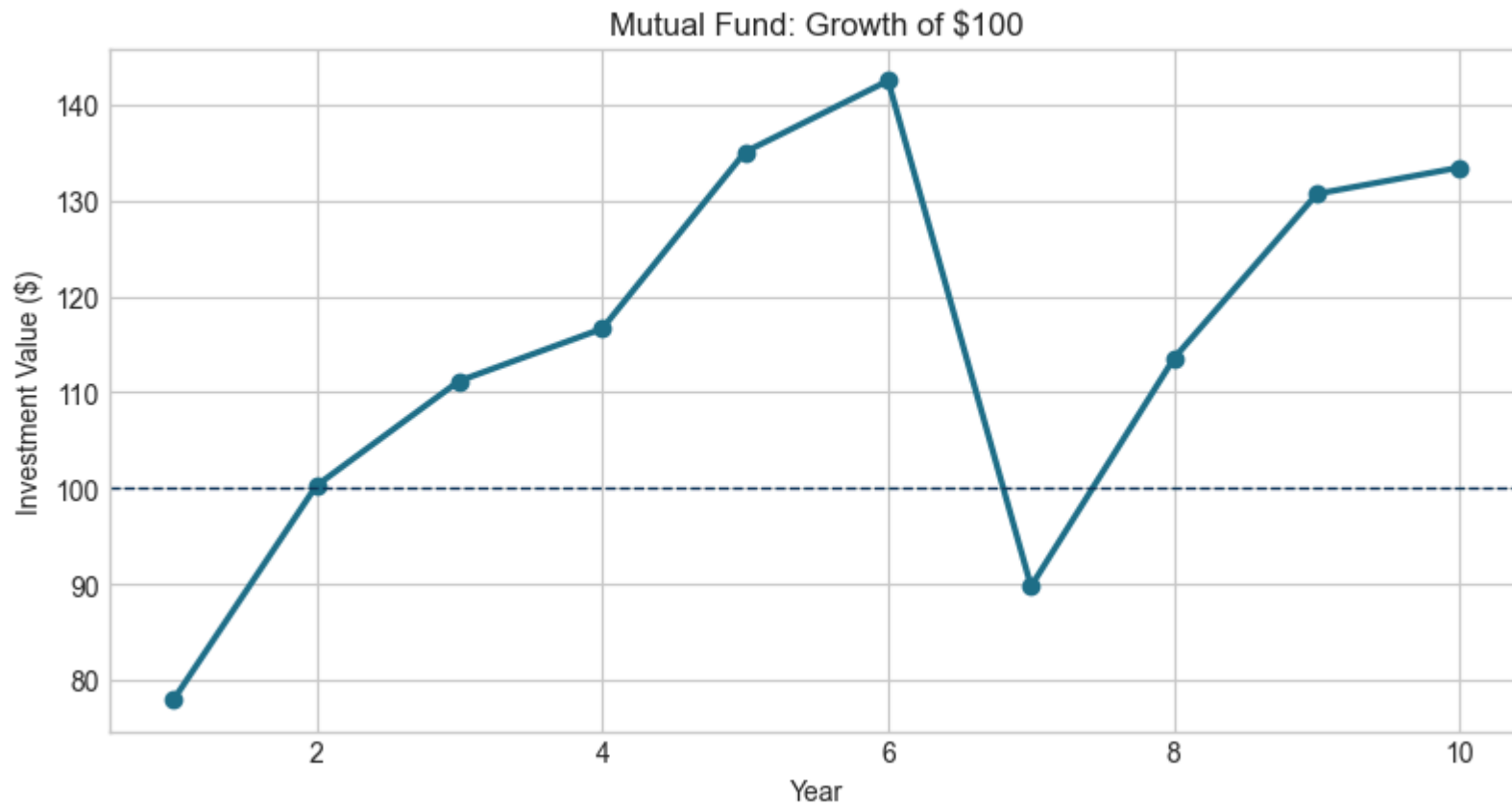
- The geometric mean is the constant rate that produces the observed compounded growth.
- It is appropriate for successive growth rates, not as a replacement for every arithmetic mean.

Geometric Mean

Arithmetic mean return: 5.04%

Geometric mean return: 2.93%

Ending value of \$100: \$133.45



Percentiles

- The pth percentile marks a position about p percent of the way through the ordered data. For example, the 25th percentile separates roughly the lowest quarter from the remaining observations.
- After sorting the data, calculate the percentile location as

$$L_p = \frac{p}{100}(n + 1)$$

- If L_p is not an integer, interpolate between the observations on either side of the location.
- A percentile is a location in an ordered distribution. It is not the percentage value of the variable.

Percentiles: Example

Data ($n = 10$): 12, 6, 20, 4, 16, 8, 22, 14, 10, 18.

Ordered position	1	2	3	4	5	6	7	8	9	10
Value	4	6	8	10	12	14	16	18	20	22

$$L_{30} = 0.30(10 + 1) = 3.3$$

Position 3.3 lies between positions 3 and 4. Interpolate 0.3 of the way from 8 to 10:

$$P_{30} = 8 + 0.3(10 - 8) = 8.6$$

$$L_{70} = 0.70(10 + 1) = 7.7$$

Position 7.7 lies between positions 7 and 8. Interpolate 0.7 of the way from 16 to 18:

$$P_{70} = 16 + 0.7(18 - 16) = 17.4$$

Quartiles

Quartiles are specific percentiles:

- $Q_1 = P_{25}$
- $Q_2 = P_{50}$, the median
- $Q_3 = P_{75}$

The quartiles are calculated using the percentile location formula from the preceding slide.

Quartiles: Example

Same data: 12, 6, 20, 4, 16, 8, 22, 14, 10, 18.

Ordered position	1	2	3	4	5	6	7	8	9	10
Value	4	6	8	10	12	14	16	18	20	22

$$L_{25} = 0.25(10 + 1) = 2.75, \quad Q_1 = 6 + 0.75(8 - 6) = 7.5$$

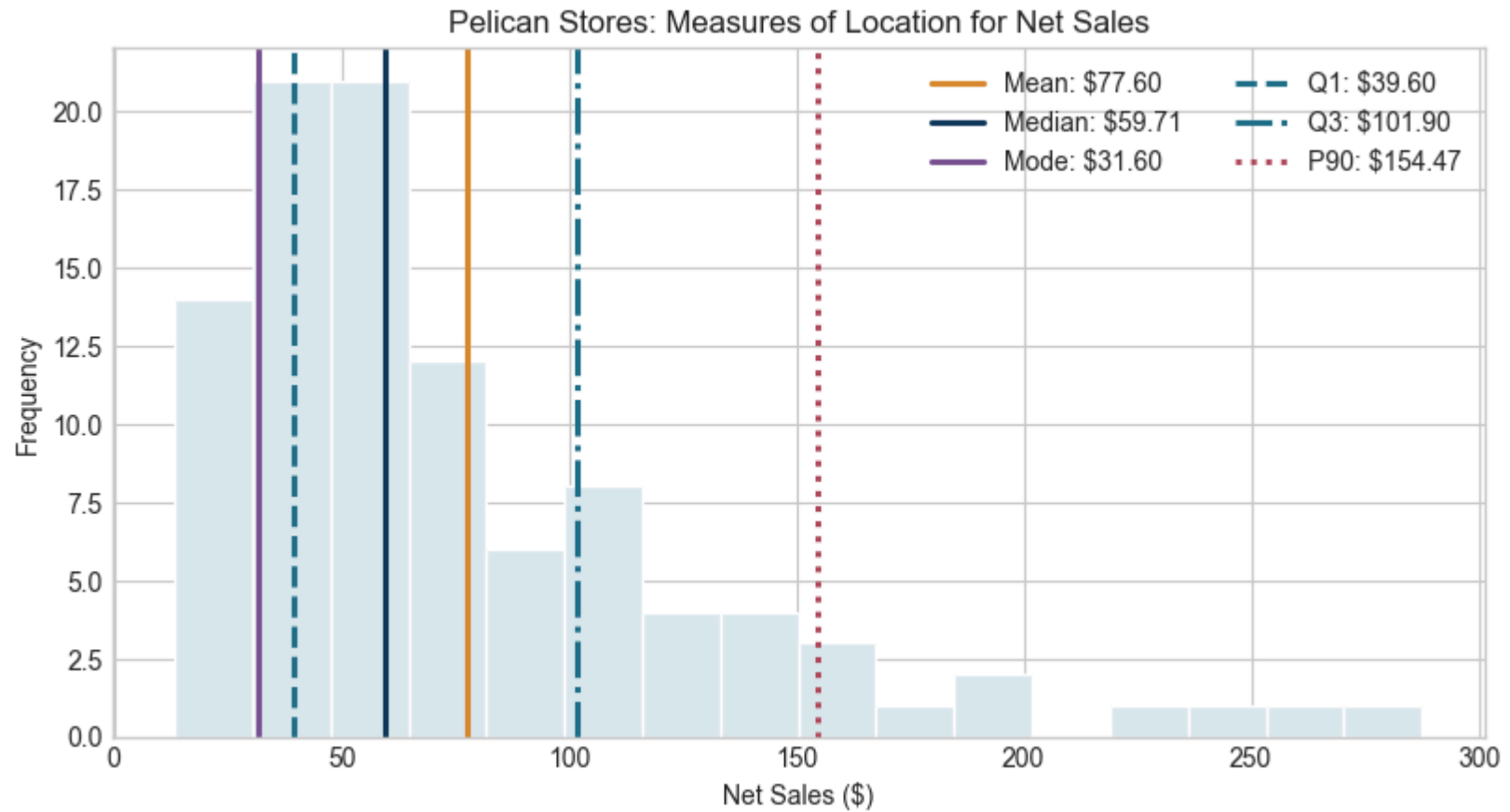
$$L_{50} = 0.50(10 + 1) = 5.5, \quad Q_2 = 12 + 0.5(14 - 12) = 13$$

$$L_{75} = 0.75(10 + 1) = 8.25, \quad Q_3 = 18 + 0.25(20 - 18) = 18.5$$

Quartiles: Example

	measure	net_sales
0	Mean	77.60
1	Median	59.71
2	Mode	31.60
3	Q1	39.60
4	Q3	101.90
5	P90	154.47

Quartiles: Example



Mean, Median, Mode, and Skewness

The mean is pulled toward the longer tail. The median usually moves less because it depends on ordered position rather than the size of every observation.

Distribution shape	Usual ordering
Negative skewness, longer left tail	Mean < Median < Mode
Symmetric and single-peaked	Mean = Median = Mode
Positive skewness, longer right tail	Mode < Median < Mean

Pelican Stores net sales follow the positive-skewness pattern:

$$\text{Mode } (\$31.60) < \text{Median } (\$59.71) < \text{Mean } (\$77.60)$$

The ordering involving the mode is a common pattern for single-peaked distributions, not a rule for every data set.

Exercise 1

A regional bank is reviewing three parts of a management report.

Customer balances

Customer	1	2	3	4	5	6
Balance, thousand MAD	2.0	2.5	3.0	3.5	4.0	80.0

Loan portfolio

Loan segment	A	B	C
Return, percent	4	6	9
Balance, million MAD	60	30	10

Deposit growth

Year	1	2	3
Growth, percent	20	-10	15

The analyst proposes using the arithmetic mean for all three parts of the report. Which measure should be used in each case, and how should each result be interpreted?

Exercise 1: Solution

- The mean customer balance is 15.83 thousand dirhams, while the median is 3.25 thousand.
 - The extreme 80 thousand balance pulls the mean upward. Report both and use the median when describing a typical observed customer.
- The portfolio yield must weight each rate by its loan balance.
 - The weighted mean is 5.1 percent. The unweighted mean of 6.33 percent gives equal importance to portfolios of different sizes.
- The geometric mean growth rate is approximately 7.49 percent per year. It is the constant annual rate that produces the same compounded three-year growth.
- The arithmetic mean answers different questions in the three settings.
- Management should not label one measure simply as the average without identifying its construction and interpretation.

Measures of Variability

- Range
- Interquartile Range
- Variance
- Standard Deviation
- Coefficient of Variation

Measures of location do not describe how observations differ from one another. Two data sets can have the same mean and very different variability.

Range and Interquartile Range

- The range uses the two most extreme observations:

$$\text{Range} = x_{\max} - x_{\min}$$

- The interquartile range covers the middle 50 percent of the ordered observations:

$$IQR = Q_3 - Q_1$$

- The range is highly sensitive to extreme values.
- The IQR is resistant but deliberately ignores variation outside the middle half.

Variance

- Variance summarizes squared deviations from the mean.

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} \quad \sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

- The sample variance uses $n - 1$ because it estimates population variability after the sample mean has been estimated from the same observations.
- Squaring prevents positive and negative deviations from canceling, but it leaves variance in squared units.

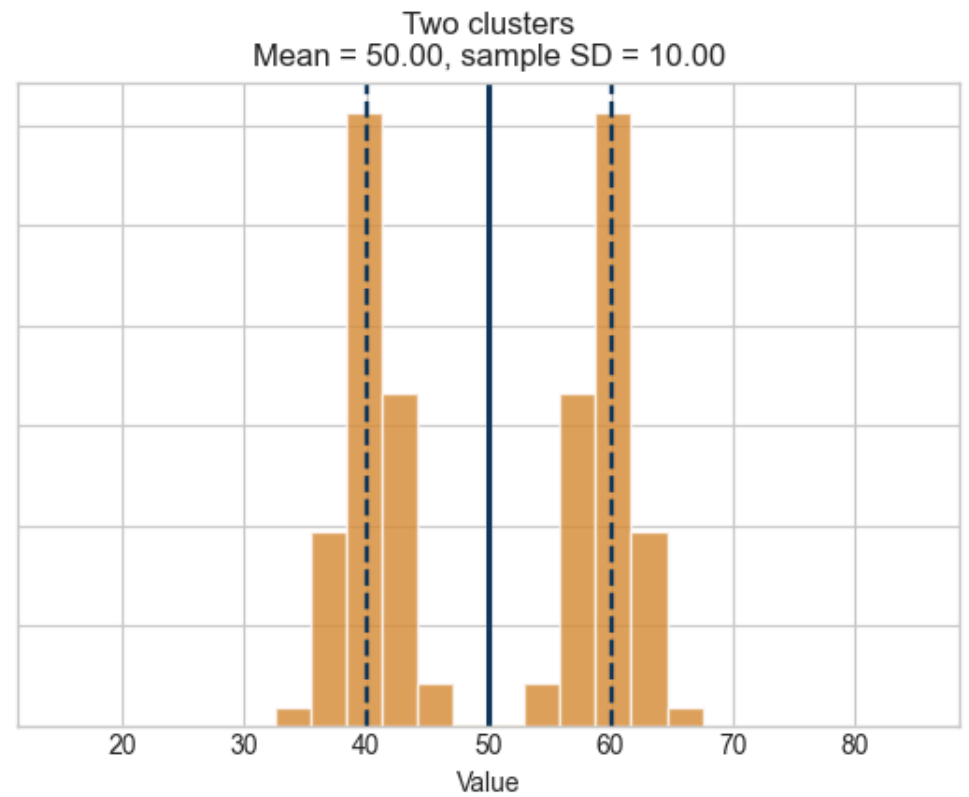
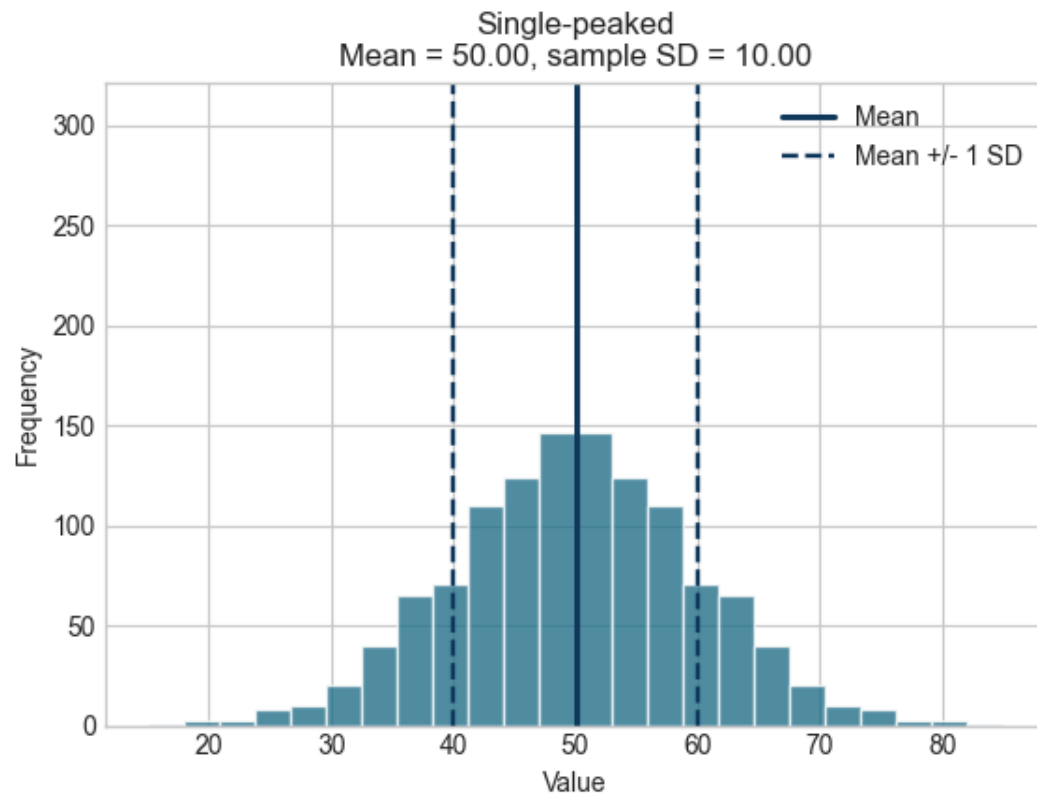
Standard Deviation

- The standard deviation is the positive square root of the variance.

$$s = \sqrt{s^2} \quad \sigma = \sqrt{\sigma^2}$$

- Standard deviation is measured in the same units as the variable.
 - A value of zero means that all observations are identical.
- A standard deviation is not an average absolute distance from the mean.
 - Its interpretation depends on the distribution and the context.

Standard Deviation



Coefficient of Variation

The coefficient of variation expresses the standard deviation relative to the mean.

$$CV = 100 \left(\frac{s}{\bar{x}} \right) \%$$

It can compare relative variability when variables have different units or substantially different means.

The coefficient of variation is meaningful for ratio-scale variables with a meaningful zero and a positive mean. It becomes unstable when the mean is close to zero and is not appropriate for arbitrary interval scales.

	mean	range	IQR	sample_ variance	sample_std_dev	population_std_dev	CV_percent
Net Sales (\$)	77.60	274.36	62.3	3098.59	55.66	55.39	71.73
Age (years)	43.08	58.00	18.0	153.49	12.39	12.33	28.76

Exercise 2

The suppliers have the following delivery-time summaries:

Measure	Supplier A	Supplier B
Mean	8 days	10 days
Median	8 days	10 days
Standard deviation	2 days	1.5 days
Range	8 days	6 days

Management prefers Supplier A because its mean is lower. Operations prefers Supplier B because its deliveries are more consistent.

Which claim is supported by the summaries? What do the measures establish, and what additional information is needed before choosing a supplier?

Exercise 2: Solution

- Supplier A is faster by both mean and median. Supplier B has the smaller standard deviation and range. The two preferences emphasize different objectives.
- $CV_A = (2/8) \times 100 = 25\%$; $CV_B = (1.5/10) \times 100 = 15\%$. Relative to its mean, B has lower variability.
- No measure makes one supplier universally better. Management must specify the cost of delay, the importance of predictability, promised delivery thresholds, and the consequences of unusually late deliveries.
- The full distributions, sample sizes, time periods, late-delivery rates, and comparable order conditions are needed. A lower average does not reveal how often a service commitment is missed.

Skewness

- Population moment skewness is

$$\gamma_1 = \frac{1}{N} \sum_{i=1}^N \left(\frac{x_i - \mu}{\sigma} \right)^3$$

- A common finite-sample adjustment is

$$G_1 = \frac{n}{(n-1)(n-2)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^3$$

- Negative values indicate a longer left tail, values near zero indicate approximate symmetry, and positive values indicate a longer right tail.
- Pelican Stores net sales have adjusted sample skewness of approximately 1.71.

Kurtosis

- Pearson, or ordinary, population kurtosis is the fourth standardized central moment:

$$\beta_2 = \frac{1}{N} \sum_{i=1}^N \left(\frac{x_i - \mu}{\sigma} \right)^4$$

- Excess kurtosis subtracts the normal-distribution reference value:

$$\gamma_2 = \beta_2 - 3$$

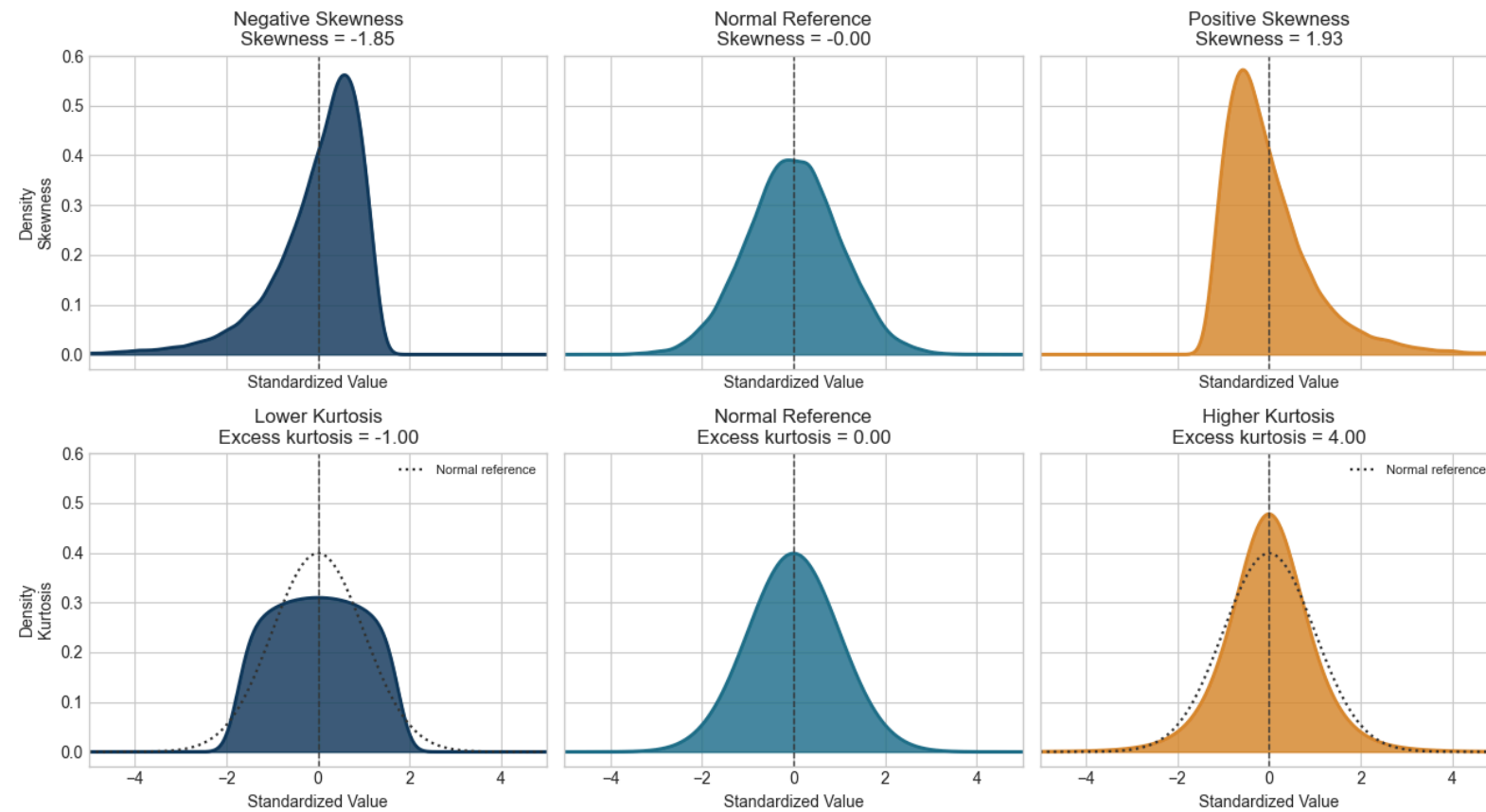
- The adjusted sample excess kurtosis is

$$G_2 = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}$$

- A normal distribution has Pearson kurtosis 3 and excess kurtosis 0.
- Positive excess kurtosis indicates greater tail extremity relative to the normal distribution; negative excess kurtosis indicates less.
- Pelican Stores net sales have adjusted excess kurtosis of approximately 3.15 and Pearson kurtosis of approximately 6.15.

Kurtosis

Distribution Shape: Skewness and Kurtosis



Direct and Adjusted Calculations

For direct sample moments, let $m_r = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^r$. Then

$$g_1 = \frac{m_3}{m_2^{3/2}}, \quad b_2 = \frac{m_4}{m_2^2}, \quad g_2 = b_2 - 3$$

The finite-sample adjustments can be written as

$$G_1 = \frac{\sqrt{n(n-1)}}{n-2} g_1, \quad G_2 = \frac{n-1}{(n-2)(n-3)} [(n+1)g_2 + 6]$$

Pelican Stores net sales	Direct calculation	Adjusted sample calculation	Normal reference
Skewness	1.689	1.715	0
Pearson kurtosis	5.935	6.150	3
Excess kurtosis	2.935	3.150	0

- Direct and adjusted calculations lead to the same shape diagnosis, but not exactly the same numerical value.
- Always state whether kurtosis is Pearson or excess and whether a sample adjustment is used.

z-Scores

- A z-score measures relative location in standard-deviation units.

$$z_i = \frac{x_i - \bar{x}}{s}$$

- A negative z-score lies below the mean.
- A positive z-score lies above the mean.
- A z-score of zero equals the mean.
- Standardization allows values from different scales to be compared by relative position.
 - It does not make differently shaped distributions equivalent.

z-Scores: Example

Using the ten observations from the percentile and quartile examples: 4, 6, 8, 10, 12, 14, 16, 18, 20, 22.

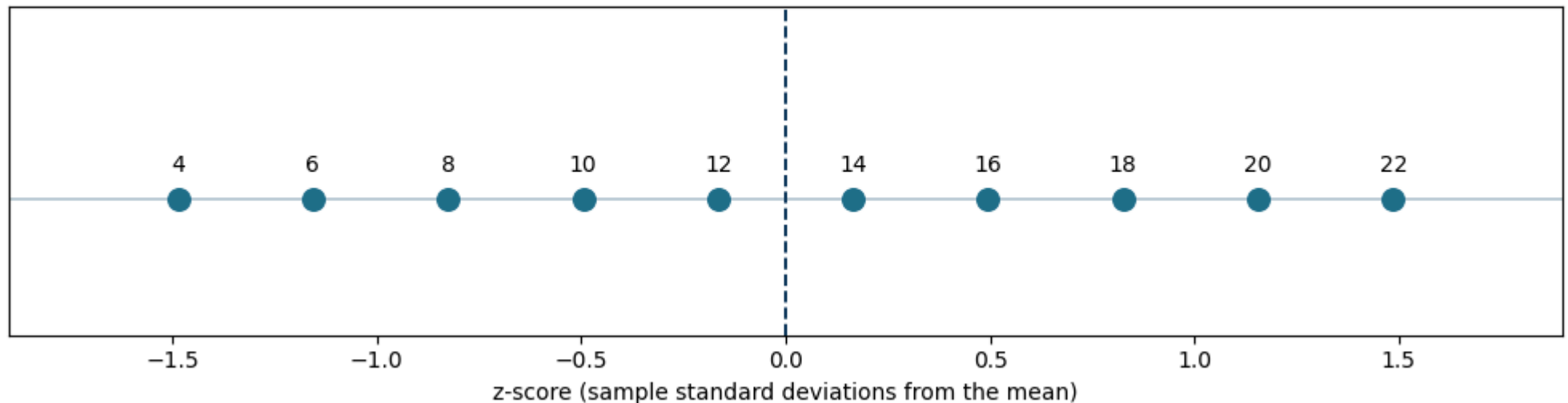
$$\bar{x} = 13, \quad \sum_{i=1}^{10} (x_i - \bar{x})^2 = 2(9^2 + 7^2 + 5^2 + 3^2 + 1^2) = 330$$

$$s = \sqrt{\frac{330}{10 - 1}} \approx 6.06$$

$$z(22) = \frac{22 - 13}{6.06} \approx 1.49, \quad z(4) = \frac{4 - 13}{6.06} \approx -1.49$$

The value 22 is about 1.49 sample standard deviations above the mean; 4 is about 1.49 below.

Ten Observations on a Standardized Scale



Chebyshev's Theorem and the Empirical Rule

Chebyshev's theorem applies to any distribution. For $k > 1$, at least

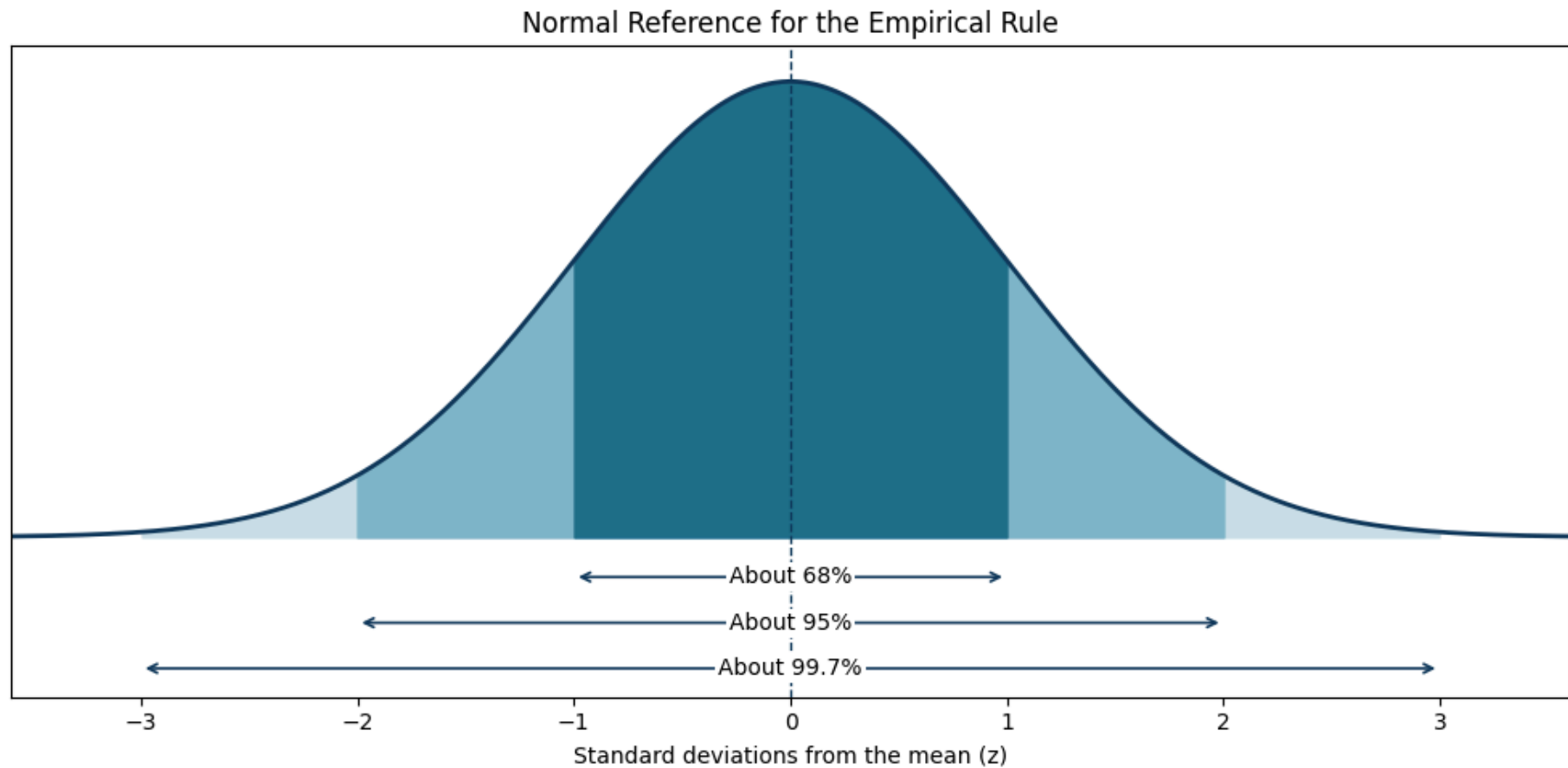
$$1 - \frac{1}{k^2}$$

of the observations lie within k standard deviations of the mean. The empirical rule requires an approximately bell-shaped distribution.

Distance from mean	Chebyshev minimum	Empirical rule
1 standard deviation	No positive guarantee	Approximately 68 percent
2 standard deviations	At least 75 percent	Approximately 95 percent
3 standard deviations	At least 88.9 percent	Approximately 99.7 percent

- Chebyshev gives a minimum guarantee.
- The empirical rule gives a more precise approximation because it imposes a shape condition.

Chebyshev's Theorem and the Empirical Rule



Detecting Outliers

- Two common rules are:
- z-score rule: flag observations with $z < -3$ or $z > 3$
- IQR rule: flag observations below $Q_1 - 1.5(IQR)$ or above $Q_3 + 1.5(IQR)$
- The rules do not necessarily identify the same observations.
 - For Pelican Stores net sales, the IQR rule based on the quartiles above flags five observations, while the z-score rule flags three.
- A flagged value may be a recording error, an observation from the wrong population, or a valid unusual case.
 - Detection requires an investigation. It does not authorize automatic deletion.

Exercise 3

An analyst applies the empirical rule to Pelican Stores net sales and reports that approximately 95 percent of customers must fall within two standard deviations of the mean.

- The analyst then removes all five observations identified by the IQR rule.
- A colleague argues that only the three observations with z-scores above 3 should be removed.

Audit both recommendations and state how the observations should be handled.

Exercise 3: Solution

- Net sales are strongly right-skewed. The bell-shape condition is not credible, so the empirical rule does not justify the 95 percent claim.
- Chebyshev's theorem applies regardless of shape and guarantees at least 75 percent within two standard deviations. It does not say that only 75 percent will be inside.
- The IQR and z-score rules use different definitions, so identifying five and three observations is not contradictory. Neither rule proves that a value is erroneous.
- Check transaction records, units, inclusion criteria, product mix, and the population definition. Valid high-value sales belong in the descriptive analysis and may be important to management.
- If corrected or excluded values are reported, the reason and the effect on the results must be documented.

Five-Number Summary

A five-number summary contains:

1. Smallest value
2. First quartile
3. Median
4. Third quartile
5. Largest value

Applying the percentile location formula above gives the following Pelican Stores net-sales summary:

Minimum	Q1	Median	Q3	Maximum
13.23	39.60	59.71	101.90	287.59

Box Plot

A box plot represents the quartiles, median, spread, and potential outliers.

- The box extends from Q_1 to Q_3 .
- A line inside the box marks the median.
- The lower and upper fences are $Q_1 - 1.5(IQR)$ and $Q_3 + 1.5(IQR)$.
- Observations beyond the fences are plotted separately as potential outliers.

A box plot summarizes a distribution but does not show every feature that a histogram or dot plot may reveal.

Comparative Box Plots

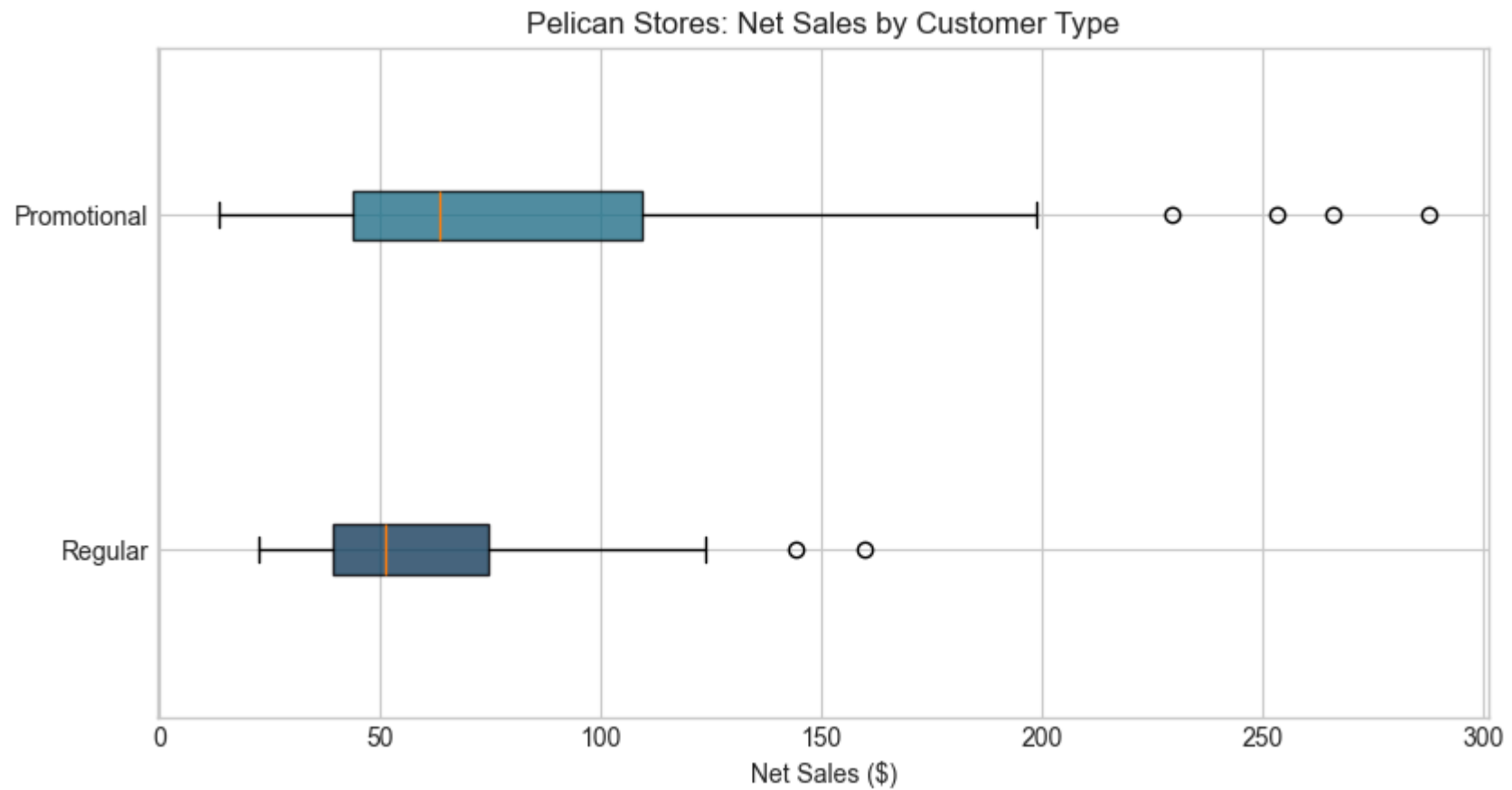
Box plots drawn on a common scale support comparisons of:

- Median location
- Middle-half variability
- Overall asymmetry
- Potential outliers

Group sizes and the percentile convention should accompany the display. A visible group difference is descriptive. It does not establish why the groups differ.

Comparative Box Plots

	N	Q1	Median	Q3	Outliers
Regular	30	39.38	51.00	74.25	2
Promotional	70	43.50	63.42	109.39	4



Exercise 4

The comparative box plots show that promotional customers have a higher median net sale and a wider middle 50 percent.

- Both customer groups also contain observations beyond their group-specific upper IQR fences.
- Management concludes that promotional status increases customer spending and proposes ending efforts directed at regular customers.

Does the display support management's conclusion?

What can be concluded from the box plots and data structure, and what further analysis is needed?

Exercise 4: Solution

- The display supports a descriptive difference in the recorded sample. Promotional customers have higher location and greater variability in net sales.
- The box plot does not show whether the samples represent the broader customer populations, whether customer types faced comparable conditions, or why they differ.
- Promotional status was not randomly assigned. Product mix, prior behavior, discount eligibility, store conditions, and customer selection may explain the difference.
- The flagged high sales should be investigated and reported, not removed because they occur in one group.
- Ending the regular-customer effort is not supported. Management needs costs, repeat purchases, margins, customer history, and a credible evaluation of the promotional intervention.

Covariance

- Covariance measures the direction of linear association between two quantitative variables.

$$s_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1} \quad \sigma_{xy} = \frac{\sum_{i=1}^N (x_i - \mu_x)(y_i - \mu_y)}{N}$$

- A positive covariance indicates that the variables tend to move in the same direction. A negative covariance indicates opposite movement.
- Its magnitude depends on the measurement units.
- Rescaling either variable changes the covariance, so magnitude alone does not provide a standardized measure of strength.

Correlation Coefficient

- Correlation standardizes covariance:

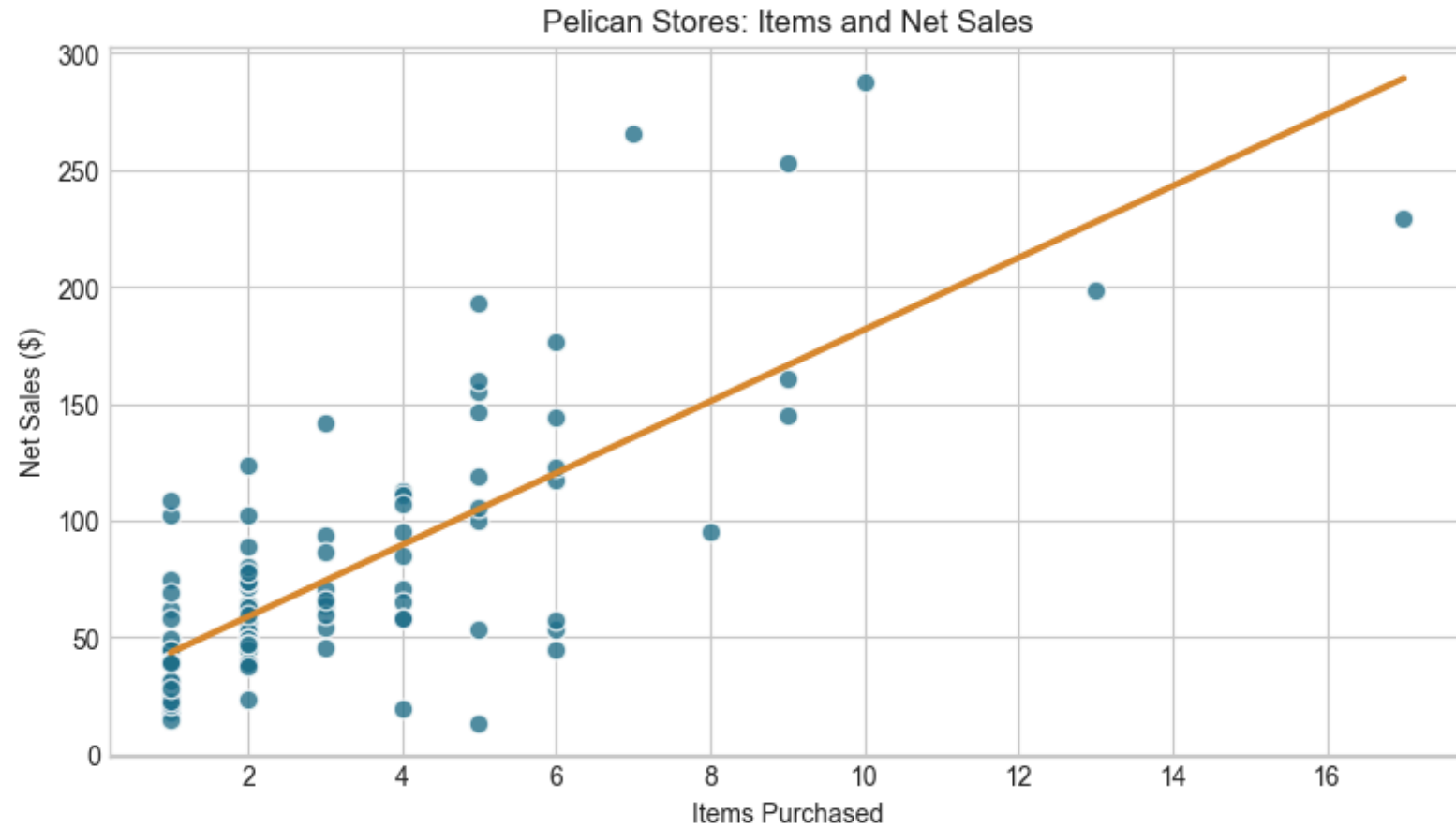
$$r_{xy} = \frac{s_{xy}}{s_x s_y} \quad \rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$$

- Correlation ranges from -1 to 1 and measures the direction and strength of linear association.
- A correlation near zero can coexist with a strong nonlinear relationship.
- Correlation can also be sensitive to unusual observations.
- High correlation does not establish causation.

Correlation Coefficient

Sample covariance: 115.00 item-dollars

Sample correlation: 0.755



Exercise 5

Five stores report the following values for two series. Both series have meaningful zero values.

Store	Series X	Series Y
A	2	8
B	4	14
C	6	6
D	8	10
E	10	12

- For each series, calculate the mean, median, sample variance, sample standard deviation, and coefficient of variation.
- Calculate the sample covariance and correlation between Series X and Series Y.

Exercise 5: Solution

Series X

$$\bar{X} = \frac{2 + 4 + 6 + 8 + 10}{5} = 6$$

Ordered values: 2, 4, 6, 8, 10, so $\text{Median}_X = 6$.

$$s_X^2 = \frac{(2 - 6)^2 + (4 - 6)^2 + (6 - 6)^2 + (8 - 6)^2 + (10 - 6)^2}{5 - 1} = \frac{40}{4} = 10$$

$$s_X = \sqrt{10} \approx 3.16 \quad CV_X = \frac{\sqrt{10}}{6}(100) \approx 52.70\%$$

Series Y

$$\bar{Y} = \frac{8 + 14 + 6 + 10 + 12}{5} = 10$$

Ordered values: 6, 8, 10, 12, 14, so $\text{Median}_Y = 10$.

$$s_Y^2 = \frac{(8 - 10)^2 + (14 - 10)^2 + (6 - 10)^2 + (10 - 10)^2 + (12 - 10)^2}{5 - 1} = \frac{40}{4} = 10$$

$$s_Y = \sqrt{10} \approx 3.16 \quad CV_Y = \frac{\sqrt{10}}{10}(100) \approx 31.62\%$$

Exercise 5: Solution

Store	X	Y	$X - 6$	$Y - 10$	$(X - 6)^2$	$(Y - 10)^2$	$(X - 6)(Y - 10)$
A	2	8	-4	-2	16	4	8
B	4	14	-2	4	4	16	-8
C	6	6	0	-4	0	16	0
D	8	10	2	0	4	0	0
E	10	12	4	2	16	4	8
Total	30	50	0	0	40	40	8

The final column contains the paired cross-products used for covariance. With $n = 5$:

$$s_{XY} = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{n - 1} = \frac{8}{4} = 2$$

Correlation standardizes the same cross-product total by the variation in both series:

$$r_{XY} = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum(X - \bar{X})^2 \sum(Y - \bar{Y})^2}} = \frac{8}{\sqrt{40(40)}} = 0.20$$

Series Y has lower variability relative to its mean. The correlation indicates a weak positive linear relationship in these five observations.

What is next?

- Introduction to Probability I
 - Random Experiments, Counting Rules, and Assigning Probabilities
 - Events and Their Probabilities
 - Some Basic Relationships of Probability
 - Reading(s):
 - SBE: Ch. 4.1 to 4.3

Exercise 6

A retailer compares two fulfillment centers:

Measure	Center North	Center South
Orders processed	9,200	2,100
Median fulfillment time	18 hours	22 hours
Mean fulfillment time	24 hours	23 hours
Standard deviation	20 hours	6 hours
IQR	11 hours	8 hours
Observations above the upper IQR fence	34	4

Management plans to move all volume to Center North because its median is lower.

- Does the lower median justify that decision?
- Compare speed, variability, unusual observations, and sample sizes, and identify any additional evidence needed.

Exercise 6: Solution

Calculation	Center North	Center South
Mean minus median	$24 - 18 = 6$ hours	$23 - 22 = 1$ hour
Coefficient of variation	$\frac{20}{24}(100) = 83.33\%$	$\frac{6}{23}(100) = 26.09\%$
Upper-fence observation rate	$\frac{34}{9,200}(100) = 0.37\%$	$\frac{4}{2,100}(100) = 0.19\%$

Median difference = $22 - 18 = 4$ hours in North's favor

Mean difference = $24 - 23 = 1$ hour in South's favor

$$\frac{s_{North}}{s_{South}} = \frac{20}{6} = 3.33 \quad \frac{IQR_{North}}{IQR_{South}} = \frac{11}{8} = 1.38$$

- North is faster at the median but slower at the mean, with substantially greater variability and a higher flagged-order rate.
- These calculations do not justify moving all volume without the full distributions, delay severity, and a capacity analysis.

Exercise 7

A manufacturer compares two packaging lines using 12 production days from each line.

Measure	Line A	Line B
Mean acceptable units per day	1,000	1,000
Median	1,010	1,080
Sample standard deviation	50	260
First quartile	975	1,010
Third quartile	1,035	1,130
Minimum	910	250
Maximum	1,080	1,180

Management prefers Line B because its median and quartiles are higher. Finance considers the lines equivalent because their means are equal. The 250-unit day for Line B coincides with a reported stoppage whose duration has not been verified.

- Which claim is supported by the summaries?
- How should the 250-unit day be handled before choosing a packaging line?

Exercise 7: Solution

Calculation	Line A	Line B
IQR	$1,035 - 975 = 60$	$1,130 - 1,010 = 120$
Coefficient of variation	$\frac{50}{1,000}(100) = 5.00\%$	$\frac{260}{1,000}(100) = 26.00\%$
Range	$1,080 - 910 = 170$	$1,180 - 250 = 930$
Median minus mean	$1,010 - 1,000 = 10$	$1,080 - 1,000 = 80$

$$\text{Median difference} = 1,080 - 1,010 = 70 \text{ units}$$

$$\frac{s_B}{s_A} = \frac{260}{50} = 5.20 \quad \frac{\text{Range}_B}{\text{Range}_A} = \frac{930}{170} = 5.47$$

For the 250-unit day on Line B:

$$z = \frac{250 - 1,000}{260} = -2.88$$

$$\text{Lower IQR fence} = 1,010 - 1.5(120) = 830 > 250$$

- Line B has higher typical output but far greater variability and an extreme low-output day.
- The 250-unit day is flagged by the IQR rule, but not by the $|z| > 3$ rule.
 - Verify the record; a genuine stoppage is part of operational risk, not a reason to discard the day. Neither claim alone justifies choosing a line.

Exercise 8

Rif Valley Capital is deciding whether to include the mutual fund represented in MutualFund.xlsx in a retirement plan. The committee's report cites only the arithmetic mean return and the best annual return. Does the available evidence support including the fund? Which numerical measures and displays should guide the decision, and what cannot be concluded from these data?

Exercise 8: Solution

Year	Return, percent	Growth factor	Squared deviation from 5.04
1	-22.1	0.779	736.580
2	28.7	1.287	559.796
3	10.9	1.109	34.340
4	4.9	1.049	0.020
5	15.8	1.158	115.778
6	5.5	1.055	0.212
7	-37.0	0.630	1,767.362
8	26.5	1.265	460.532
9	15.1	1.151	101.204
10	2.1	1.021	8.644
Total or product	50.4	1.3345	3,784.464

$$\bar{r} = \frac{50.4}{10} = 5.04\%$$

$$\text{Median} = \frac{5.5 + 10.9}{2} = 8.20\%$$

$$s^2 = \frac{3,784.464}{10 - 1} = 420.496 \quad s = \sqrt{420.496} = 20.506 \text{ percentage points}$$

$$\text{Range} = 28.7 - (-37.0) = 65.7 \text{ percentage points}$$

Exercise 8: Solution

Using $L_p = (p/100)(n + 1)$:

$$L_{25} = 2.75, \quad Q_1 = -22.1 + 0.75[2.1 - (-22.1)] = -3.95\%$$

$$L_{75} = 8.25, \quad Q_3 = 15.8 + 0.25(26.5 - 15.8) = 18.48\%$$

$$IQR = 18.48 - (-3.95) = 22.43 \text{ percentage points}$$

The IQR fences are -37.59% and 52.11% , so no annual return is flagged by the IQR rule.

$$\bar{r}_G = (1.3345)^{1/10} - 1 = 2.93\% \quad 100(1.3345) = \$133.45$$

The geometric mean is below the arithmetic mean because gains and losses compound. Ten annual observations, without a benchmark or fees, are insufficient for deciding whether the fund belongs in a retirement plan.